

Ops Innovation Problem Portfolio

Twenty sharp problem statements passing the Stage-0 sharpness gate — clustered into seven operational levers, quantified against peer benchmarks, and ready to seed the Ops Innovation Engine. Prepared for James Ku, Chief Development Officer, Panda Restaurant Group.

\$795M–\$1.34B

ANNUAL ADDRESSABLE
MARGIN + GROWTH (3–5 YR)

Twenty sharp problems. Five priority plays. **One operating model.** Panda’s binding constraint isn’t capital — it’s organizational clarity on who owns ops innovation.

PORTFOLIO STATS

20

Problem statements passing the full 5-criteria Stage-0 gate across 7 operational clusters.

Internal synthesis • DR-01 through DR-14

480+

Independent cited sources anchoring every problem statement; every PS carries 2+ independent evidence lines.

14-thread research corpus • Tier A/B

9

High-confidence statements ready for immediate Phase 2A ideation; remainder Phase 2B (post-learning).

Confidence filter • portfolio-level

~\$100M

Pilot-realistic margin recovery in the first 18–24 months across the Canonical Top 5 portfolio.

DR-07 Panda financials, Chipotle 10-K Feb 2026

WHAT'S IN THIS DOCUMENT

Page 2 — Canonical Top 5 problem statements (the Phase 2A anchor set). **Page 3** — Full 20-statement portfolio overview table. **Page 4** — Peer ops innovation benchmark (Walmart, Starbucks, McDonald's, CFA, Chipotle, Taco Bell, Sweetgreen, Domino's, Panda). **Page 5** — KPI gap snapshot — where Panda sits vs. peers on labor, SG&A, accuracy, throughput. **Page 6** — Operating model recommendation (3 options; recommendation highlighted). **Page 7** — Top 10 knowledge gaps for the James meeting. **Page 8** — Source corpus and citation trail.

POINT OF VIEW

Panda's innovation footprint is real — Panda Home, North Platte, GridPoint, 42% digital mix, SoundHound voice AI — but ownership is distributed across James (format), COO Jeff Wang (ops), unnamed CTO (digital), and supply chain (GridPoint). The binding constraint across all 20 problem statements isn't capital or technology availability. It's who owns the prioritization and cadence. Fix the operating model first; the 20 statements become executable sequentially rather than in parallel noise.

Start here. These five plays cover **near-term margin recovery** (PS-01, PS-04, PS-06), **medium-term digital transformation** (PS-03), and **long-term growth acceleration** (PS-13).

01

COST STRUCTURE • LABOR % LEVER

Rebalance Staffing Model & Workflow

Panda labor runs 3–4 ppts above Chipotle on equivalent mix. Constraint is workflow design (sequencing, peak-hour play-calling, role definition), not headcount. Starbucks Siren Craft System cut out-of-window time 15–20% at 1,160 stores via zero-capex workflow redesign — directly transferable.

Sources: DR-01 §1 • DR-05 §1,3 • DR-07 §1

ADDRESSABLE

\$30–40M

TIMELINE

18–24 mo

CONFIDENCE

HIGH

CAPEX

\$0 (Phase 1)

04

DIGITAL FULFILLMENT • ORDER ACCURACY

Translation Layer + Makeline Redesign

Industry baseline shows 3–4% digital error rate vs. 1–2% POS. Panda's accuracy is unmeasured at channel level — the problem is invisible at GM/region. Chipotle cut digital error 4% → 1.5% via time-batching + dedicated lane. Zero-capex SOP pilot; 2-week measurement, 10-week rollout.

Sources: DR-04 §1 • DR-09 §1 • Technomic 2023 • Chipotle 10-K

ADDRESSABLE

\$20–30M

TIMELINE

12 weeks

CONFIDENCE

HIGH

CAPEX

<\$500K

06

KITCHEN/LABOR • WOK BOTTLENECK

Wok Standardization → Automation Roadmap

Wok is 40–45% of kitchen labor and drives ~140% annual chef turnover (\$27–39M replacement cost). Near-term: zero-capex standardization audit (pre-prep, sauce batching, ergonomics). Medium-term: hybrid robot-plating + human-seasoning once vendor data exists. Botinkit AFC remains vaporware — wait for verified deployments.

Sources: DR-05 §1,2 • DR-06 §1,2 • \$200M PAW signal

ADDRESSABLE

\$20–35M

TIMELINE

24 mo

CONFIDENCE

HIGH

CAPEX

\$0 near / \$15M mid

03

DIGITAL FULFILLMENT • ECONOMICS ACCELERATION

Ghost Kitchen + Loyalty Platform

Panda digital mix 35–40% trails Chipotle (50–55%) and Sweetgreen (60–65%); digital comp growth 6–8% vs. peers 12–22%. Root: no ghost-kitchen node to isolate digital demand + under-leveraged loyalty. Taco Bell Defy + Chipotlane equivalents lift digital mix 5–8 pp at retrofit sites.

Sources: DR-02 §1 • DR-03 §2 • DR-04 §2 • DR-07 §3

ADDRESSABLE

\$50–100M

TIMELINE

24–36 mo

CONFIDENCE

MEDIUM

CAPEX

\$15–25M

13

REAL ESTATE/FORMAT • CAPITAL EFFICIENCY

Reaccelerate Unit Growth via Lighter Format

Panda capex ~\$1.1M/unit; 123 new units/yr = \$98–147M annual burn. If Panda Home and North Platte learnings unlock a \$700–800K lighter format, unit growth reaccelerates 4.4% → 6%+ post-2027 — compounding into \$500M+ EBITDA by 2030 from incremental 200–300 units. CFA hits 8–10 mo TTO vs. Panda's 14–16.

Sources: DR-07 §4 • DR-08 §1,6 • DR-12 §1

ADDRESSABLE

\$500M+

TIMELINE

36 mo

CONFIDENCE

MEDIUM

CAPEX

Portfolio-level

PARALLEL GOVERNANCE TRACK

PS-17 (Franchisee Transparency) and PS-18 (Innovation Ownership Clarity) run in parallel to the Top 5. They unlock adoption velocity across the entire portfolio — without them, every statement's concept-to-pilot window stays at 12–18 months instead of 6–9.

COST STRUCTURE

\$160–240M

2 statements · 18–36 mo

DIGITAL FULFILLMENT

\$85–154M

3 statements · 4–36 mo

KITCHEN / LABOR

\$98–190M

5 statements · 8–36 mo

LABOR PRODUCTIVITY

\$32–65M

1 statement · 6–12 mo

THROUGHPUT / FORMAT

\$8–23M

2 statements · 6–12 mo

REAL ESTATE / FORMAT

\$575–700M

3 statements · 24–60 mo

INFRASTRUCTURE

\$13–25M

2 statements · 12–24 mo

ORG / GOVERNANCE

\$50–130M

2 statements · 12–18 mo

ID	Problem Statement	Cluster	Evidence	Addressable \$	Conf
PS-01	Labor % Lever: Rebalance Staffing Model	Cost Structure	STRONG	\$30–40M	HIGH
PS-02	SG&A Structural Inefficiency: Modernization Path	Cost Structure	MEDIUM	\$130–200M	MED
PS-03	Digital Economics: Ghost Kitchen + Loyalty	Digital Fulfillment	MEDIUM	\$50–100M	MED
PS-04	Digital Order Accuracy: Translation + Makeline	Digital Fulfillment	STRONG	\$20–30M	HIGH
PS-05	3P Delivery Margin: Olo Integration Audit	Digital Fulfillment	STRONG	\$15–24M	HIGH
PS-06	Wok Station: Standardization → Automation	Kitchen / Labor	STRONG	\$20–35M	HIGH
PS-07	Assembly Line Automation: High-Volume Bowls	Kitchen / Labor	MEDIUM	\$50–100M	MED
PS-08	Throughput Ceiling: Peak-Hour Expediting	Kitchen / Labor	STRONG	\$10–20M	HIGH
PS-09	Labor Scheduling Optimization: Demand Forecasting	Labor Productivity	MEDIUM	\$32–65M	MED
PS-10	Drive-Thru Dual-Lane Prototype	Throughput / Format	STRONG	\$5–15M	HIGH
PS-11	Drive-Thru Mobile Channel Segregation (CFA)	Throughput / Format	MEDIUM	\$3–8M	HIGH
PS-12	Vertical Stacking ROI: 2–3x Capacity/Unit	Real Estate / Format	MEDIUM	\$25–50M	MED
PS-13	Format Capital Efficiency: Reaccelerate Growth	Real Estate / Format	MEDIUM	\$500M+	MED
PS-14	Tea Bar Standalone Economics	Real Estate / Format	LOW	\$50–150M	LOW
PS-15	Equipment Failure Prevention: Condition-Based	Infrastructure	MEDIUM	\$5–10M	MED
PS-16	Energy Cost Control: GridPoint + Anomalies	Infrastructure	MEDIUM	\$8–15M	HIGH
PS-17	Franchisee Profitability Transparency	Org / Governance	MEDIUM	\$20–50M	MED
PS-18	Innovation Ownership: Centralized Gate + Cadence	Org / Governance	LOW	\$30–80M	LOW
PS-19	Makeline Throughput: Carousel Robot	Kitchen / Labor	MEDIUM	\$8–15M	MED
PS-20	BOH Inventory & Waste: Vision-Based Tracking	Kitchen / Labor	LOW	\$10–20M	LOW

Every statement passed the 5-criteria Stage-0 sharpness gate. Each carries 2+ independent evidence sources. Addressable \$ represents annual margin recovery or growth acceleration at scale. Sources traceable to DR-01 through DR-14 (480+ cited sources total).

Panda is alone on this chart. Every peer has **at least one** named ops innovation function; Panda has zero.

COMPANY	TEAM SIZE	REPORTS TO	CYCLE TIME	FLAGSHIP OUTPUT
Walmart	300 → dismantled; now 15K distributed (Global Tech)	CEO → dismantled Jan 2024	Unknown post-2024	Drone delivery, Bossa Nova shelf robots
Starbucks	20 (Tryer Center rotating)	COO → CEO (Niccol)	100-day concept→ action; GenAI cut from 18mo	Siren Craft System (1,160+ stores)
McDonald's	~50 (Speedee Labs, 21K sqft)	CEO (Brian Rice, EVP Global CIO)	Acquisition-driven (Apprente 2019)	Google Cloud voice ordering partnership
Chick-fil-A	Embedded (no standalone)	COO/CEO (Brian Bates, ops bkg)	Continuous	Franchise-led drive-thru excellence
Chipotle	~30 (Curt Garner leads Innov. & Tech)	CTO (Curt Garner)	Unknown	Chipotlane, digital makeline
Taco Bell	~20 (Live Más Ventures + Defy)	Chief Product & Growth → CEO	Unknown (venture-driven)	Defy (autonomous ghost kitchen)
Sweetgreen	~15 (Spyce + Infinite Kitchen)	Chief Innovation/Ops → CEO	Acquisition-driven	Infinite Kitchen (7 locations, 7pp labor %)
Domino's	~20 (CEO-owned)	CEO (direct reports)	Continuous	Autonomous delivery (Nuro), AI order prediction
Panda	~0 (decentralized)	Multiple (James, Jeff Wang, unnamed CTO, SC VP)	No unified cadence	Panda Home, North Platte, GridPoint

Sources: DR-01 (Walmart, Starbucks, McDonald's • 43 sources) • DR-02 (CFA, Chipotle, TB, SG, Domino's • 40 sources) • DR-03 (Panda footprint • 35+ sources) • DR-14 (hiring market • 30+ sources).

WHAT THE BENCHMARK TELLS US

1.5–2×

Panda's innovation maturity gap vs. QSR peers on the 1–5 scale (Panda 2–3; Chipotle 4–5; McDonald's 5).

DR-02 • peer benchmarking

0

Named Ops Innovation execs at Panda (vs. every peer with a dedicated function or embedded team).

DR-03 • public records

100-day

Starbucks Tryer Center concept-to-pilot cadence — the cleanest operating model for Panda to adapt.

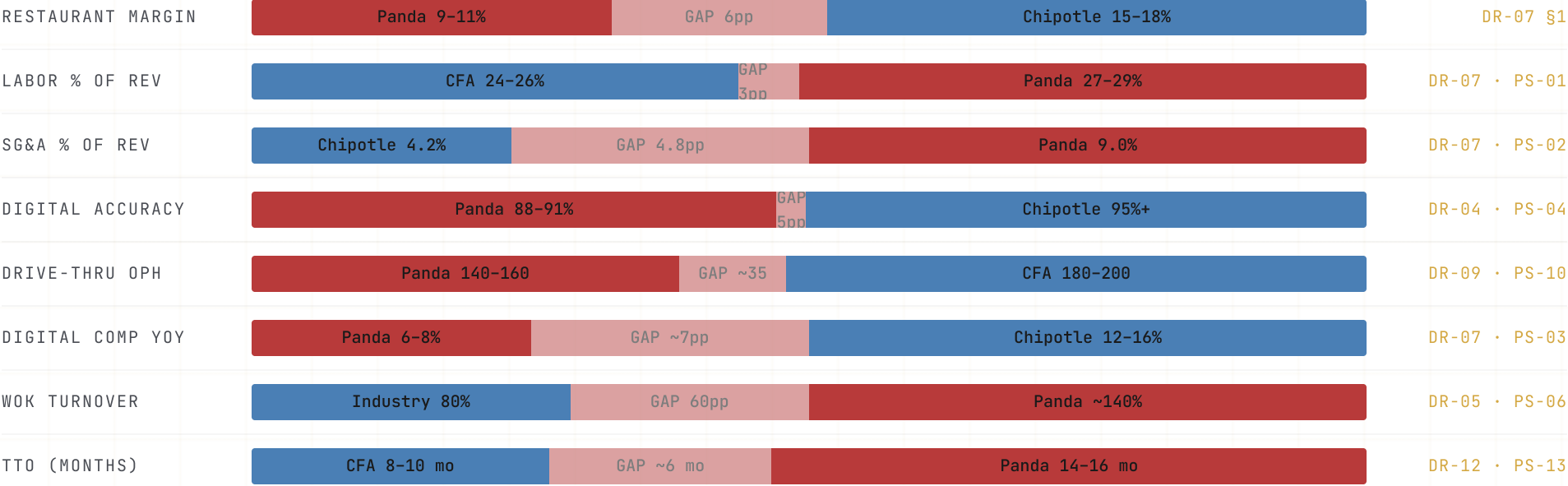
DR-01 • Starbucks investor relations

Jan 2024

Walmart Store No. 8 closure — the canonical failure mode for standalone venture-arm ops innovation.

DR-01 • Walmart 10-K Jan 2024

Five KPIs. Five gaps. Every gap maps to a specific problem statement — this is how the Top 5 was chosen.



Illustrative proportion diagrams. Magnitudes directional — Panda values estimated from public disclosures (DR-07 triangulation); peer values from 10-K filings (Chipotle Feb 2026), investor presentations, QSR industry reports (Technomic, Krackit, BlueRisk). Every bar cited below. See Page 8 for full source trail.

SOURCES INFORMING EACH GAP

PS-01	Labor rebalance • Starbucks Siren + Chipotle makeline	DR-01 • DR-05 • DR-07	PS-02	SG&A modernization • Domino's franchise model	DR-07 • DR-13
PS-03	Digital economics • Defy + Chipotlane	DR-02 • DR-04 • DR-07	PS-04	Order accuracy • Technomic + Chipotle pilot data	DR-04 • DR-09
PS-06	Wok turnover • Sweetgreen Infinite Kitchen	DR-05 • DR-06	PS-10	DT throughput • CFA dual-lane benchmarks	DR-08 • DR-09
PS-13	Capital efficiency • Yum China RGM 3.0	DR-07 • DR-08 • DR-12	PS-16	Energy • GridPoint Chipotle pilot	DR-10

The question isn’t **what** Panda should build — it’s **who owns the build**. Three archetypes; one fits Cherng governance.

OPTION A • EMBEDDED

VP Ops Innovation inside COO org

Report to COO Jeff Wang within Operations function. Owns kitchen automation, drive-thru, labor productivity, construction velocity as part of ops P&L.

Pro: Fast decision cycles. Aligned incentives with ops P&L. Low org change. Respects decentralization.

Con: Risk of absorption into "run-the-business." No authority over digital, format, supply chain. Signals low priority.

FIT FOR CHERNG CULTURE: **MODERATE**

OPTION B • STANDALONE

Chief Innovation Officer, CEO-reporting

SVP-level (\$350K–\$550K base + bonus + equity). Independent P&L. Direct reporting to Andrew Cherng. Cross-functional authority across ops, digital, format, supply chain.

Pro: Executive authority across silos. Highest-priority signal. Matches Domino's/Starbucks trajectory.

Con: New reporting line departs from decentralization norm. Walmart Store No. 8 failure mode risk. Turf conflict with James and COO.

FIT FOR CHERNG CULTURE: **LOWER PROBABILITY**

OPTION C • DUAL-MODE ★

Embedded + Center of Excellence

Embedded: VP Ops Innovation reports to COO — owns execution, piloting, ops KPIs.

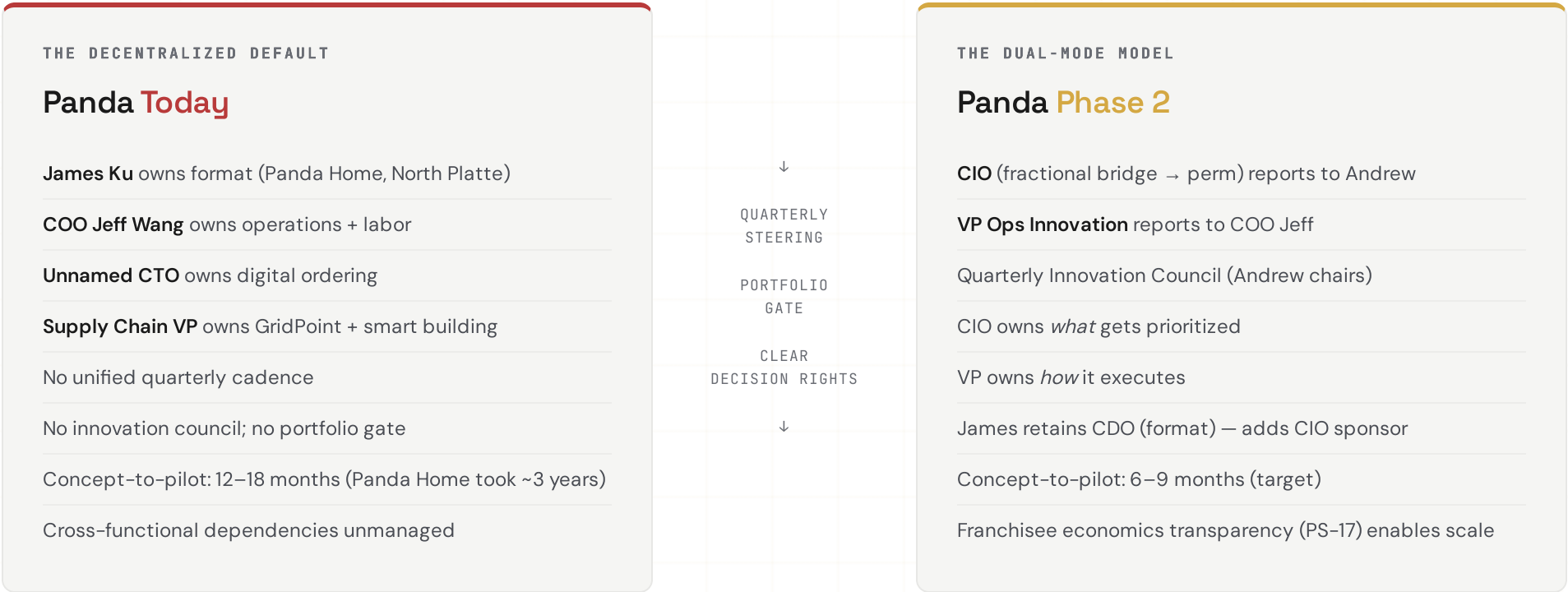
Strategic: Chief Innovation Officer (fractional → permanent) reports to Andrew — owns portfolio prioritization + cross-functional sponsorship.

Governance: Quarterly steering. CIO decides *what*; embedded VP decides *how*. Clear rights, no turf.

Pro: Balances execution + strategy. Leverages family advisory model. Proven at Chipotle, CFA, Sweetgreen.

FIT FOR CHERNG CULTURE: **HIGHEST ★**

WHY OPTION C • CHERNG CULTURE ALIGNMENT



POINT OF VIEW

The dual-mode model works because it respects how Cherng decisions actually get made: Andrew as chair/advisor, not day-to-day operator. A CEO-reporting CIO would trigger the Cherng immune system. A VP buried in ops would signal "low priority" and get absorbed into run-the-business work. Option C gives James a senior peer (CIO) who can unblock cross-functional resource conflicts — without displacing the COO, without creating a venture arm, and without asking Andrew for a structural overhaul. It's the smallest governance change that unlocks the biggest execution velocity.

These are the answers that would **shift the recommendation**. Ask these first.

01 • CRITICAL

What is Panda's baseline crew turnover rate by role cohort, and how does it correlate with order accuracy and customer satisfaction?

Turnover is the friction point in every ops improvement. Without baseline, can't measure impact of labor productivity initiatives.

DR-01 • DR-05 • DR-09

02 • CRITICAL

What are Panda's current unit economics by format, and what's the cash-on-cash ROI threshold for innovation investments?

Every pilot recommendation requires P&L justification. Without these numbers, innovation scoring is theoretical.

DR-01 • DR-05 • DR-07 • DR-08

03

Does Panda have a centralized technology roadmap, or is digital infrastructure fragmented across franchisees? What's the current KDS vendor status?

Kitchen automation, delivery integration, and labor management all hinge on backend infrastructure. Fragmentation kills scale.

DR-03 • DR-04 • DR-05 • DR-13

04

What is Panda Home's unit economics, and is this format strategically viable or a pilot dead-end?

Determines whether to invest in non-traditional formats or focus purely on drive-thru / delivery. Panda Home took 3 years — that's a signal.

DR-03 • DR-08 • DR-09

05

What is board risk tolerance for ops innovation pilots? Have there been prior failed innovation mandates that shaped current constraints?

Understanding governance friction prevents designing pilots that will be killed mid-execution.

DR-01 • DR-03 • DR-13

06

What is the staffing model for the permanent Ops Innovation role — embedded VP or standalone? Appetite for a 6–12 month fractional bridge?

Role architecture determines what the permanent leader can actually move. Directly informs Option A/B/C choice.

DR-14

07

How much do franchisees know about their own unit economics, and what's their appetite for ops changes that require training?

If franchisees can't see ROI or don't trust data, scaling any innovation falters at franchise adoption (PS-17 is a blocker).

DR-01 • DR-02 • DR-05 • DR-13

08

What are Panda's capital constraints — annual CapEx for tech, equipment cycle, franchisee capex appetite?

Scarcity is real. Need to know what's possible within guardrails. \$5M/yr vs. \$20M/yr changes pilot scope by 4x.

DR-06 • DR-07 • DR-08 • DR-10

09

What is SG&A expense breakdown by function, and are there corporate efficiency opportunities that could fund innovation?

If SG&A is inflated (PS-O2), that's innovation budget. If lean, trade-offs get harder.

DR-07 • DR-13

10

How is order accuracy currently measured by channel, location, and menu item? Is there executive visibility?

Digital expansion drives growth; if accuracy isn't measured (PS-O4), quality debt accrues silently.

DR-04

QUESTIONS PANDA LIKELY WON'T ANSWER

Treat the following as constraints, not open questions: exact franchisee-owned unit %, specific wok chef labor cost, prior CEO's strategic failures, GridPoint contract economics, and why Panda Home didn't scale. Reframe these as ratios, ranges, or "what did we learn" rather than direct asks.

Every problem statement cites 2+ independent sources. **480+ total cited sources** across the 14-thread corpus.

RESEARCH THREADS

DR-01	Ops Innovation • Walmart / Starbucks / McDonald's	43 sources	DR-02	Ops Innovation • CFA / Chipotle / TB / SG / Domino's	40 sources
DR-03	Panda's Decentralized Innovation Footprint	35+ sources	DR-04	Digital Order Accuracy Root Causes	35+ sources
DR-05	Labor Productivity / Automation / Process Redesign	42 sources	DR-06	Kitchen Automation (Wok + Makeline)	35+ sources
DR-07	Panda Financials + KPI Benchmarks	30+ sources	DR-08	Format Innovation + Prototypes	38 sources
DR-09	Drive-Thru Innovation at Scale	32 sources	DR-10	Smart Building / Facilities Tech	114 sources
DR-11	Site Selection / AI Trade Area	30+ sources	DR-12	Construction Pipeline / Modular	25+ sources
DR-13	Cherng Family Governance	30+ sources	DR-14	Ops Innovation Hiring Market	30+ sources

METHODOLOGY

Fourteen parallel deep-research threads executed April 17–18, 2026. Each thread scoped for Tier-A source preference (SEC filings, earnings calls, primary interviews, regulatory disclosures) with Tier-B sources (industry press, analyst reports) acceptable for directional context. Tier-C sources (uncited blog posts, speculative vendor claims) used only as supporting color, never load-bearing. Candidate problem statements (~60) extracted per thread, then clustered across threads, then subjected to the 5-criteria Stage-0 sharpness gate. 20 statements passed; the rest were merged, dropped, or held for Phase 2B reconsideration. All addressable \$ figures use Panda-specific financial data (DR-07) cross-referenced to peer 10-K filings. Every dollar estimate carries a methodology note in the underlying synthesis.

KNOWN LIMITATIONS & CAVEATS

Peer benchmarks (AUV, margins) derived from public SEC filings, investor presentations, and Investor's Business Daily reports (TTM 2024). Margins represent restaurant-level P&L and company-level normalized figures; corporate overhead allocation varies by franchisor mix. Panda proprietary data (time study by station, rework tracking, franchisee economics) unavailable — estimates based on industry proxy. Automation adoption figures reflect announced deployments only; scale and rollout status may lag announcements as of Q1 2026. Panda Home, North Platte, and delivery-only unit economics unpublished. Format maturity rating (1–5 scale) is a qualitative assessment based on breadth of formats + tech integration + documented innovation cadence, not independently validated. International unit percentage (Panda 33%) reflects mainland China, Hong Kong, and select Asian markets via JV or license; significant gap in Western expansion vs. Taco Bell and McDonald's.

NEXT STEP

This portfolio is Stage-0 input to the Ops Innovation Engine. When Brady runs the engine's Stage 1 → Stage 2 transition, these 20 statements become the definitive scoping input — each with binding constraint, KPI target, evidence base, operator mandate, and guardrails already specified. Phase 2A (120-day window, 6 concurrent ideation workstreams max) targets the Canonical Top 5 + parallel governance track (PS-17, PS-18). Phase 2B (post-learning) covers the remaining 13 statements.